MQTT Firewall Intrusion Detection Report

Generated: 2026-05-03 10:53:30
Source file: `JSONL (887 verdicts) + .log (887 verdicts)`
Time range: 10:42:43 → 10:53:29

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1. Executive Summary

The LightGBM-based MQTT firewall processed **4,531 packets** (5,686 log lines) and issued **893 flow-level verdicts**. Of these, **893 (100.0%) were classified as attacks** and **0 (0.0%) as benign**. Mean model confidence on attack flows was **0.9974**.

Metric	Value
Total log lines	5,686
Total packets intercepted	4,531
Total flow verdicts	893
Attack verdicts	893 (100.0%)
Benign verdicts	0 (0.0%)
Observation window	10:42:43 → 10:53:29

2. Attack Type Breakdown

Attack Type	Verdict Count	Share (%)
MQTT-DoS-Connect_Flood	796	89.1%
Recon-OS_Scan	28	3.1%
MQTT-DDoS-Connect_Flood	16	1.8%
TCP_IP-DDoS-SYN	13	1.5%
MQTT-DDoS-Publish_Flood	13	1.5%
TCP_IP-DDoS-UDP1	12	1.3%
MQTT-DoS-Publish_Flood	8	0.9%
TCP_IP-DDoS-TCP	4	0.4%
ARP_Spoofing	2	0.2%
Recon-Port_Scan	1	0.1%

3. Model Confidence Statistics

Statistics computed over **attack-classified flows only**.

Statistic	Value
Sample size	893 flows
Minimum confidence	0.6199
Maximum confidence	1.0000
Mean confidence	0.9974
Flows with confidence ≥ 0.90	880 (98.5%)

4. Flow Statistics

Metric	Min	Max	Mean
Packets per flow	3	10	4.6
Bytes per flow	0	31,712	236
Flow duration (s)	0.0004	95.9162	5.5514

5. Timing Performance — Per Attack Type

> **infer** = `model.predict()` only | **classify** = feature extraction + inference + decision | **inspect** = packet arrival → verdict (inline classifications only; 0 for background reaper-triggered flows)

Attack Type	N	Infer Mean (ms)	Infer Std (ms)	Classify Mean (ms)	Classify Std (ms)	Inspect Mean (ms)	Inspect Std (ms)
ARP_Spoofing	2	1.391	0.891	2.157	0.886	0.000	0.000
MQTT-DDoS-Connect_Flood	16	0.968	1.022	1.958	1.565	2.521	1.609
MQTT-DDoS-Publish_Flood	13	0.699	0.806	1.467	1.362	1.850	1.378
MQTT-DoS-Connect_Flood	796	0.396	0.232	0.871	0.444	1.151	0.613
MQTT-DoS-Publish_Flood	8	0.474	0.050	1.189	0.209	1.379	0.222
Recon-OS_Scan	28	0.762	0.897	1.439	1.336	1.571	1.324

Attack Type	N	Infer Mean (ms)	Infer Std (ms)	Classify Mean (ms)	Classify Std (ms)	Inspect Mean (ms)	Inspect Std (ms)
Recon-Port_Scan	1	4.391	0.000	7.977	0.000	8.207	0.000
TCP_IP-DDoS-SYN	13	0.689	0.705	1.491	1.159	1.713	1.204
TCP_IP-DDoS-TCP	4	0.438	0.072	0.963	0.199	1.286	0.215
TCP_IP-DDoS-UDP1	12	0.461	0.094	1.161	0.377	1.164	0.256
Overall	893	0.435	0.377	0.944	0.652	1.201	0.741

6. Top Attacker Flows

265 distinct `(src_ip:src_port, attack_type)` flows detected.

Source (ip:port)	Classified Attack Type	Verdict Count
127.0.0.1:1883	MQTT-DoS-Connect_Flood	608
127.0.0.1:1883	MQTT-DDoS-Connect_Flood	16
127.0.0.1:1883	TCP_IP-DDoS-TCP	4
127.0.0.1:56319	MQTT-DoS-Connect_Flood	2
127.0.0.1:46415	MQTT-DoS-Connect_Flood	2
127.0.0.1:1883	MQTT-DoS-Publish_Flood	2
127.0.0.1:49864	MQTT-DoS-Connect_Flood	1
127.0.0.1:56039	TCP_IP-DDoS-SYN	1
10.255.255.254:34690	TCP_IP-DDoS-UDP1	1
10.255.255.254:49859	ARP_Spoofing	1
10.255.255.254:38523	TCP_IP-DDoS-UDP1	1
127.0.0.1:43109	TCP_IP-DDoS-SYN	1
127.0.0.1:44089	TCP_IP-DDoS-SYN	1
127.0.0.1:59273	MQTT-DDoS-Publish_Flood	1
127.0.0.1:54611	MQTT-DDoS-Publish_Flood	1
127.0.0.1:48951	MQTT-DDoS-Publish_Flood	1
127.0.0.1:58563	MQTT-DDoS-Publish_Flood	1

Source (ip:port)	Classified Attack Type	Verdict Count
127.0.0.1:42561	TCP_IP-DDoS-SYN	1
127.0.0.1:56951	MQTT-DDoS-Publish_Flood	1
10.255.255.254:41499	TCP_IP-DDoS-UDP1	1

...and 245 additional flows not shown.

7. Last 10 Verdicts

Time	Verdict	Attack Type	Confidence	Source	Destination	Pkts	Byte
10:52:05	ATTACK	MQTT-DDoS-Connect_Flood	1.0000	127.0.0.1:1883	127.0.0.1:47041	8	31,7
10:52:20	ATTACK	MQTT-DDoS-Connect_Flood	1.0000	127.0.0.1:1883	127.0.0.1:38869	8	40
10:53:16	ATTACK	MQTT-DoS-Connect_Flood	1.0000	127.0.0.1:1883	127.0.0.1:54395	3	10
10:53:16	ATTACK	MQTT-DDoS-Connect_Flood	1.0000	127.0.0.1:1883	127.0.0.1:47041	3	9,041
10:53:22	ATTACK	MQTT-DoS-Connect_Flood	1.0000	127.0.0.1:1883	127.0.0.1:56203	3	0
10:53:22	ATTACK	MQTT-DoS-Connect_Flood	1.0000	127.0.0.1:1883	127.0.0.1:34063	3	0
10:53:23	ATTACK	MQTT-DoS-Connect_Flood	1.0000	127.0.0.1:1883	127.0.0.1:34063	3	0
10:53:23	ATTACK	MQTT-DoS-Connect_Flood	1.0000	127.0.0.1:1883	127.0.0.1:56203	3	0
10:53:28	ATTACK	MQTT-DoS-Connect_Flood	1.0000	127.0.0.1:1883	127.0.0.1:38869	3	10
10:53:29	ATTACK	TCP_IP-DDoS-SYN	0.9954	127.0.0.1:53901	127.0.0.1:1883	10	86

Report generated by `scripts/parse_firewall_logs.py` — MQTT IDS/IPS Benchmarking Framework.